

Supplementary information

Spectral concentration in two large Vina docking-score matrices: two-way centering reveals structured residual variation without a resolved target-ranking gain

Adeliya Leleytner, Victor Safronov, Maxim Fedorov

Supplementary figures

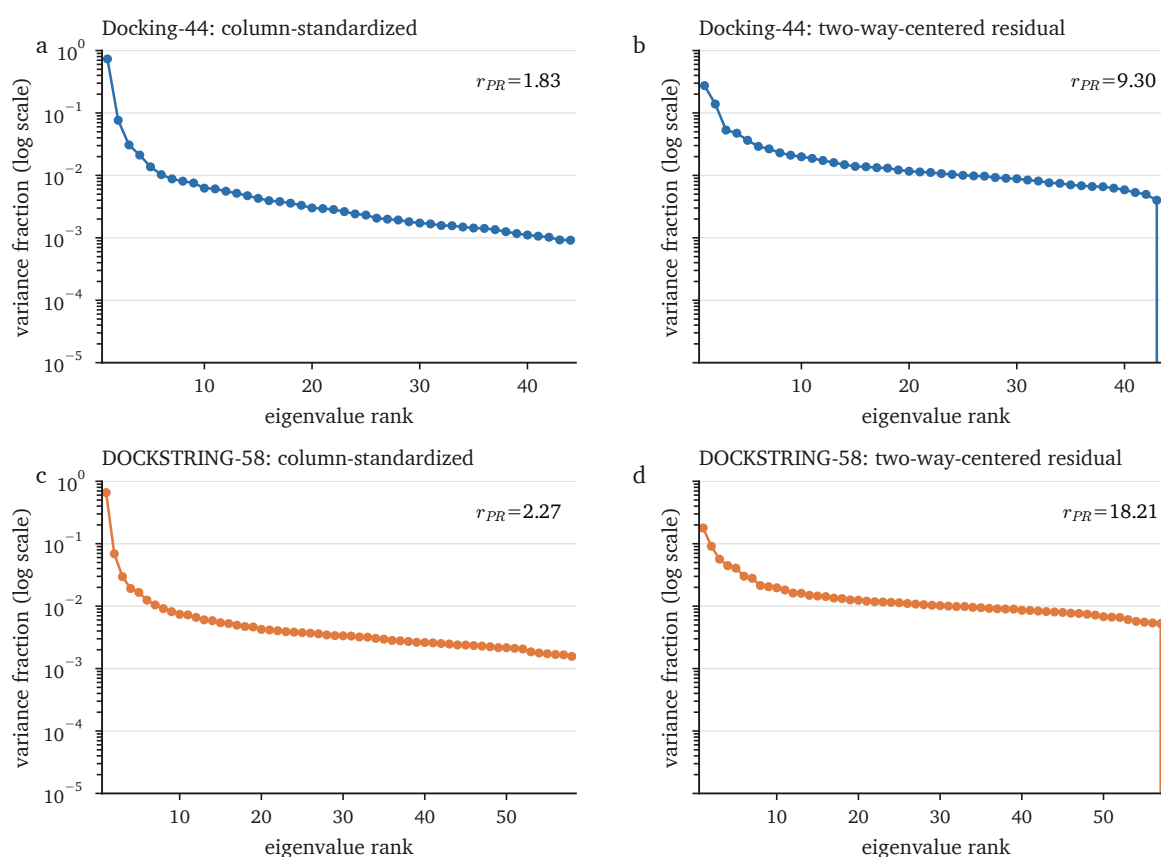


Figure S1: Full correlation-spectrum scree plots. **a,b**, Docking-44 column-standardized and two-way-centered residual spectra. **c,d**, DOCKSTRING-58 column-standardized and residual spectra. Eigenvalues are divided by the number of targets so that each spectrum sums to one. The ordinate is logarithmic. Double centering imposes one exact zero eigenvalue; values at zero fall below the plotting range.

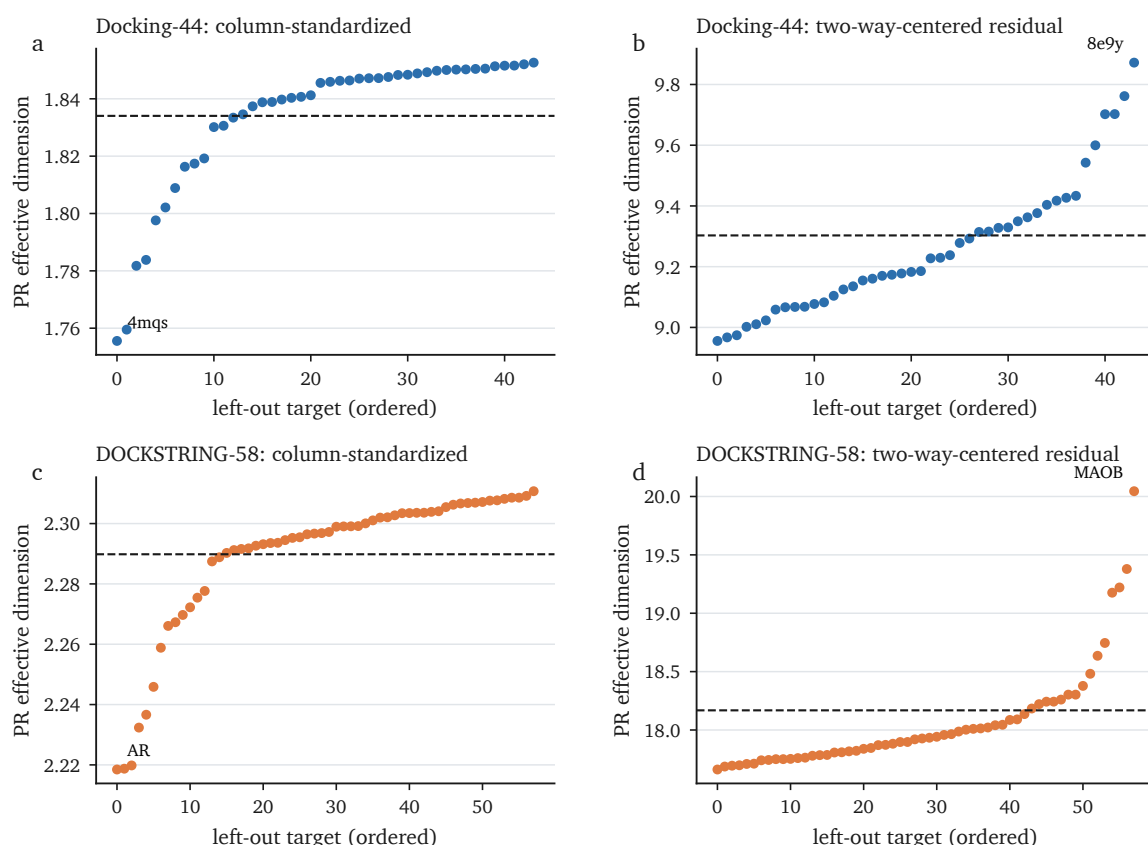


Figure S2: Leave-one-target-out composition sensitivity. Each point is the PR effective dimension after deleting one target; targets are ordered by the resulting value within each panel. Dashed lines mark the complete-panel estimate on the same ligand support. Labels identify the most influential deletion. These ranges describe the two observed target panels and are not confidence intervals for a target superpopulation.

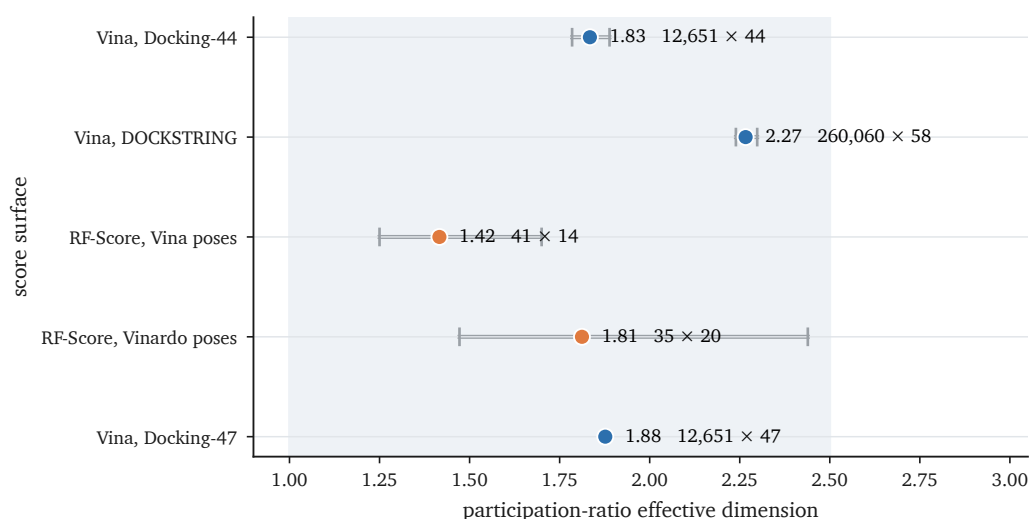


Figure S3: Illustrative scoring-function and target-expansion sensitivities. PR effective dimensions and source-analysis bootstrap intervals are shown for the two large Vina panels, RF-Score v1 on Vina poses, RF-Score v1 on Vinardo poses and the Docking-47 expansion. Point labels report ligand and target counts. The 47-target expansion is a deterministic sensitivity and is shown without an interval. The RF-Score blocks contain only 35–41 ligands and are not large independent replications.

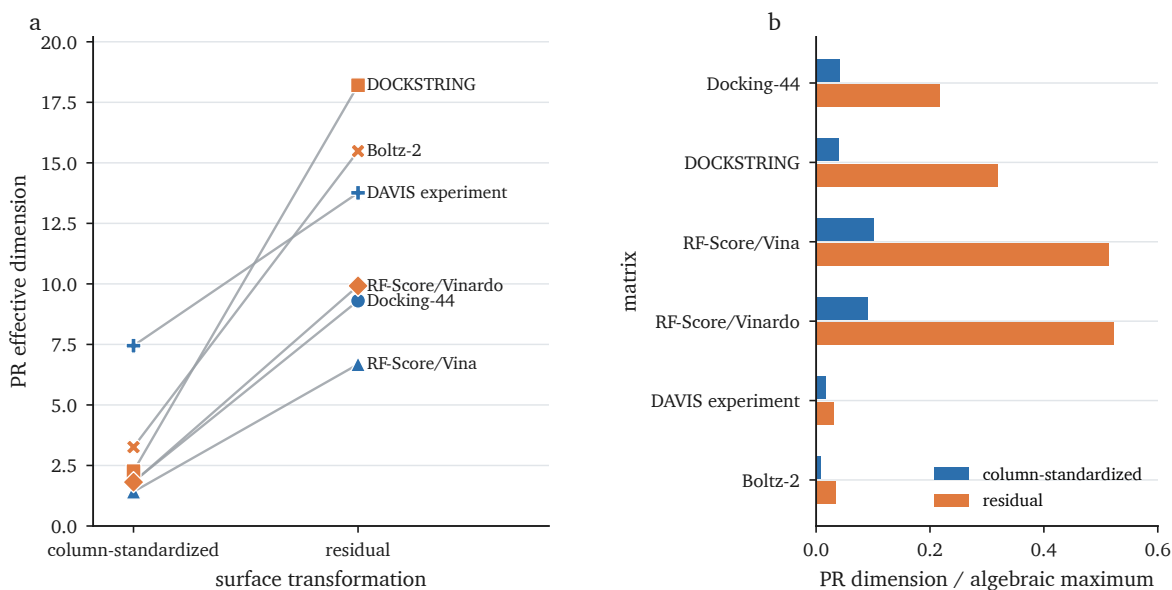


Figure S4: Contextual centering ladder across score and affinity surfaces. **a**, Column-standardized and two-way-centered residual PR effective dimensions for the two Vina matrices, two small RF-Score blocks, experimental DAVIS affinity and Boltz-2. Lines connect the same matrix under the two transformations. **b**, Each value divided by its algebraic maximum, P before and $P - 1$ after centering. The learned-affinity examples are contextual sensitivities, not docking replications or accuracy benchmarks.

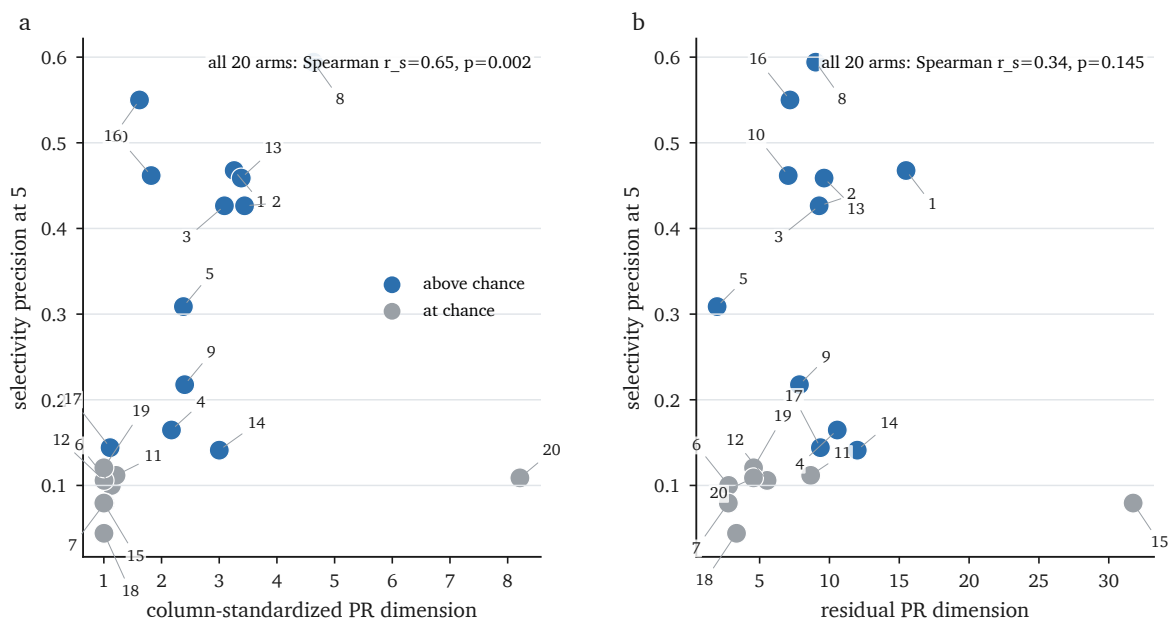


Figure S5: Exploratory DTI association on the complete 20-arm panel. Selectivity precision at five is plotted against the column-standardized PR effective dimension (**a**) and the two-way-centered residual PR effective dimension (**b**). Numbers identify arms in Table S6; blue denotes arms above the outcome-related chance criterion and grey denotes arms at chance. Correlations are descriptive because the arms share data, architectures and representations.

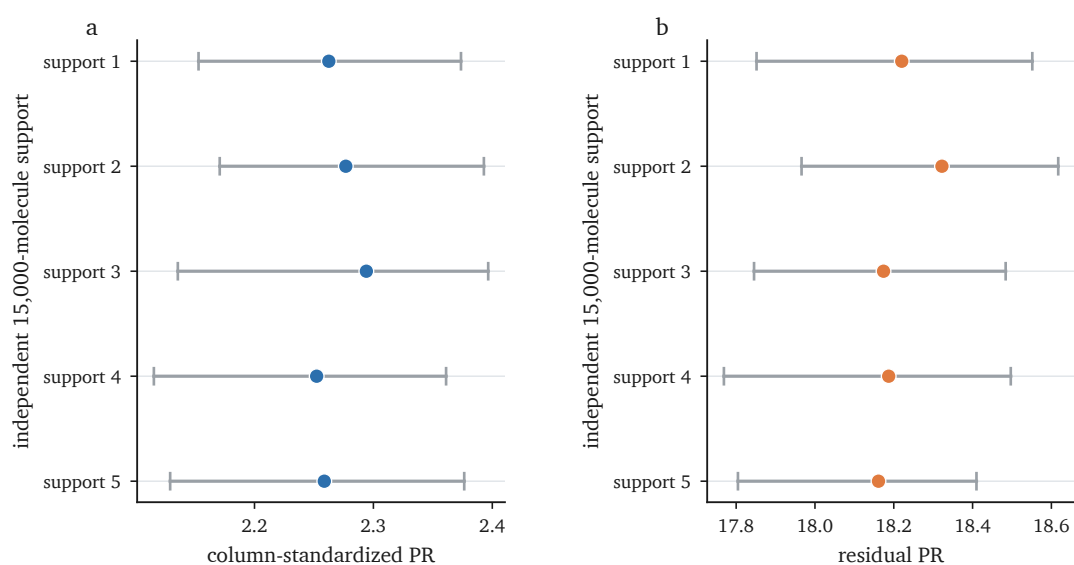


Figure S6: DOCKSTRING chemical-support and scaffold sensitivity. Each row is an independently seeded 15,000-molecule support. Points are support-specific PR dimensions and horizontal intervals are 95% Bemis–Murcko scaffold-cluster bootstrap intervals from 100 replicates. **a**, Column-standardized surface; **b**, two-way- centered residual. All estimates are conditional on the same 58 targets.

Table S1: Dataset and preprocessing summary. PR denotes participation-ratio effective dimension. Listed procedures quantify sensitivity within the observed chemical and target panels, not superpopulation uncertainty.

Matrix	scorer / source	<i>N</i>	<i>P</i>	missing	preprocessing	column-standardized PR	residual PR	entropy rank	uncertainty
Docking-44	Vina-GPU 2.0	12,651	44	1.466%	positive/source-censored scores to 0; target-mean imputation; column z-score	1.834	9.303	3.897	Butina-cluster bootstrap; target leave-one-out and subsampling
DOCKSTRING-58	AutoDock Vina	260,060	58	0.002%	positive scores to 0; complete rows; column z-score	2.266	18.206	6.206	scaffold-cluster bootstrap on five independent chemical supports; target jackknife
RF-Score/Vina-pose sensitivity	RF-Score v1	41	14	0.000%	complete common block; column z-score	1.416	6.681	2.158	ligand bootstrap; small fixed panel
RF-Score/Vinardo-pose sensitivity	RF-Score v1	35	20	0.000%	complete common block; column z-score	1.812	9.918	3.347	ligand bootstrap; small fixed panel
Matched docking	AutoDock Vina	74	6	0.000%	complete matched block; column z-score	1.827	3.646	2.605	same ligand-target support as exact-relation median ChEMBL block
Matched ChEMBL affinity	experimental pChEMBL	74	6	9.910%	exact relation; median over repeats; target-median imputation; column z-score	3.833	4.128	4.628	same observed support; assay and imputation sensitivities

Table S2: Docking-44 preprocessing and censoring sensitivity. Complete-case and target-exclusion results are not directly comparable because they change chemical or target support.

Surface	preprocessing	PR	note
Column standardized	target-mean imputation (primary)	1.834	same 12,651 ligands
Column standardized	target-median imputation	1.835	same support
Column standardized	complete cases	1.320	9,297 ligands; support changes
Column standardized	primary mean-imputed matrix restricted to complete rows	1.320	same 9,297-row support; equals complete case
Column standardized	pairwise-complete Pearson correlation	1.788	no matrix imputation
Column standardized	pairwise Spearman correlation	1.686	rank correlation
Column standardized	censored zeros treated as missing	1.829	111 cells; target-mean imputation
Column standardized	exclude ligands with censored zeros	1.814	12,544 ligands
Column standardized	exclude four most-censored targets	1.602	changes the target estimand
Residual	center then standardize (primary)	9.303	closed-form two-way centering
Residual	standardize then center	9.378	order sensitivity
Residual	robust scale then center	9.462	median/IQR scaling

Table S3: Experimental-assay, missing-data and repeatability controls. The noise scale was 0.493 pChEMBL, estimated from different-document repeats.

Analysis block	N	P	PR	note
ChEMBL all endpoints, frozen refetch A	212	22	6.846	already human-only; KNN; 52.9% missing
human binding Ki/Kd	175	19	5.248	assay type B; KNN
IC50/EC50	88	15	3.921	KNN
fully observed sub-block	31	6	3.608	no imputation
matched exact-relation median	74	6	3.833	same-support docking 1.83
matched human binding, all endpoints	47	6	3.510	same-support docking 1.87
matched human binding Ki/Kd	35	6	3.072	same-support docking 1.89
ChEMBL all endpoints, frozen refetch B, KNN5	212	22	5.721	different frozen extraction; 52.9% missing
212 x 22, MICE	212	22	8.592	mean across imputations
212 x 22, SoftImpute	212	22	2.644	low-rank matrix completion
212 x 22, PPCA-EM	212	22	7.283	probabilistic PCA
matched docking + calibrated noise	74	6	2.407	95% simulation interval 2.21–2.64
matched docking + 2.5x calibrated noise	74	6	4.175	approximately reaches experimental raw PR
matched docking, linear descriptor residual	74	6	2.806	seven RDKit descriptors; same support
matched docking, cross-fitted nonlinear residual	74	6	2.680	five-fold HistGBM; same support
matched docking residual, exact-relation support	74	6	3.646	two-way centered
matched experimental residual, exact relation median	74	6	4.128	two-way centered; primary curation
matched experimental raw, 100 MICE draws	74	6	3.823	95% across imputations 3.57–4.02
matched experimental residual, 100 MICE draws	74	6	4.069	95% across imputations 3.83–4.21
paired raw experiment-docking contrast	74	6	2.006	ligand-bootstrap interval 1.17–2.67
paired residual experiment-docking contrast	74	6	0.483	ligand-bootstrap interval -0.07–0.92

Table S4: Target-panel and chemical-support sensitivities. Leave-one-target-out values describe composition sensitivity within each observed panel; they are not superpopulation confidence intervals.

Matrix	surface	full PR	LOTO min	LOTO median	LOTO max	target
Docking-44	column-standardized	1.834	1.756	1.846	1.853	4mqs
Docking-44	two-way-centered residual	9.303	8.956	9.206	9.872	8e9y
DOCKSTRING-58	column-standardized	2.266	2.218	2.297	2.311	AR
DOCKSTRING-58	two-way-centered residual	18.206	17.662	17.931	20.045	MAOB

Family-stratified target subsampling is shown in Fig. 1c,d. The descriptive rank-wise and family-wise-error-controlled simultaneous mode counts were both 3 versus 7 for Docking-44 and 4 versus 10 for DOCKSTRING-58. Across 5 independent 15,000-molecule DOCKSTRING supports, raw point estimates ranged from 2.252 to 2.294 and residual estimates from 18.161 to 18.322; support-specific scaffold intervals are plotted in Fig. S6. The fitted additive null gave median residual PR dimensions 42.40 and 56.48, compared with observed values 9.29 and 18.16. The empirical residual-permutation null gave medians 42.40 and 56.48. Across 200 matched one-ligand-per-Butina-cluster Docking-44 supports, the observed median was 9.43 versus 42.33 under permutation.

Table S5: Expanded observed-pair ChEMBL target-preference benchmark. The primary support comprises exact-relation median pChEMBL cells for 137 Docking-44 ligands and 38 targets. Accuracy averages within-ligand pairwise accuracies so that highly profiled ligands do not dominate; predicted score ties receive half credit. Intervals resample Bemis–Murcko scaffold clusters; the cohort experimental prior is fitted with the evaluation ligand’s entire scaffold cluster held out. Identity shuffles are reported only for score-based representations.

Representation	accuracy	scaffold 95% CI	pair-weighted	pairs	identity-shuffle mean / <i>p</i>
absolute Vina	0.566	0.502–0.623	0.584	2,522	0.519 / 0.014
docking target prior	0.525	0.456–0.591	0.585	2,522	–
external ChEMBL target prior	0.525	0.472–0.579	0.565	2,522	–
cohort experimental target prior (scaffold held out)	0.592	0.531–0.654	0.644	2,522	–
column-standardized Vina	0.546	0.485–0.604	0.513	2,522	0.499 / 0.048
two-way residual Vina	0.564	0.501–0.626	0.532	2,522	0.508 / 0.024

The residual-minus-column difference was +0.017 with a scaffold-bootstrap interval -0.022–0.054; its conservative Murcko/Butina 90% interval was -0.015–0.049. The residual-minus-absolute difference was -0.002 (-0.065–0.062); its conservative 90% interval was -0.056–0.051. Post hoc equivalence was not established at ± 0.02 for either contrast or at ± 0.05 for residual versus absolute; residual versus column-standardized met only the liberal ± 0.05 criterion. Unscaled target-centered and two-way-centered representations were rank-identical at accuracy 0.556 (difference +0.000); residual-target scaling alone gave 0.549. On fixed support and same-endpoint pairs, absolute, column-standardized and residual accuracies were 0.553, 0.538 and 0.555; the residual-minus-absolute cluster-interval union was -0.060–0.064. On human binding Ki/Kd, residual accuracy was 0.579 and residual minus absolute was -0.052–0.085.

Table S6: All 20 exploratory DTI scorer arms. Plot IDs identify points in Fig. S5. The above-chance flag is outcome-related and is not used to define this complete panel.

ID	arm	above chance	affinity r	column-standardized PR	residual PR	P@5	AUPRC
1	boltz_s3	yes	0.483	3.26	15.48	0.468	0.424
2	s1_readout	yes	0.553	3.44	9.30	0.426	0.423
3	chembert_davisooof	yes	0.609	3.09	9.26	0.426	0.379
4	chembert_kiba2davis	yes	0.154	2.17	10.56	0.165	0.208
5	complex_kiba2davis	yes	0.378	2.38	1.96	0.309	0.311
6	deepdta_kiba2davis	no	0.072	1.13	2.78	0.100	0.161
7	dmpnn_kiba2davis	no	0.218	1.02	2.76	0.079	0.162
8	esm_target_davisooof	yes	0.744	4.63	9.01	0.594	0.545
9	esm_target_kiba2davis	yes	0.313	2.40	7.85	0.218	0.240
10	histgbm_davisooof	yes	0.617	1.82	7.05	0.462	0.412
11	histgbm_kiba2davis	no	0.196	1.21	8.65	0.112	0.178
12	kronrls_davisooof	no	0.279	1.00	5.53	0.106	0.143
13	mlp_davisooof	yes	0.618	3.38	9.62	0.459	0.393
14	mlp_kiba2davis	yes	0.152	3.00	11.99	0.141	0.203
15	morgan_cnn_kiba2davis	no	0.241	1.00	31.75	0.079	0.153
16	rf_davisooof	yes	0.638	1.62	7.17	0.550	0.460
17	rf_kiba2davis	yes	0.217	1.11	9.35	0.144	0.189
18	ridge_davisooof	no	0.304	1.00	3.34	0.044	0.116
19	ridge_kiba2davis	no	0.146	1.00	4.57	0.121	0.141
20	transformer_cnn_kiba2davis	no	0.083	8.21	4.56	0.109	0.151

Table S7: Provenance for the outcome-restricted 12-arm DTI sensitivity panel. DAVIS OOF denotes six-fold GroupKFold by target; KIBA transfer denotes zero-shot evaluation on DAVIS after KIBA training. Metrics are listed for all arms in Table S6.

ID	arm	representation / head	training and evaluation	caveat
1	Boltz-2 deployed affinity head	co-folded complex; deployed affinity score	foundation-model training; no DAVIS fitting in this analysis; external zero-shot DAVIS	foundation-model training corpus may contain related structures or affinities
2	Boltz-2 frozen representation + ridge	frozen pre-pairformer representation; RidgeCV	DAVIS training folds; six-fold unseen-target GroupKFold	shares Boltz-2 representation with boltz_s3
3	ChemBERTa + AAC, DAVIS OOF	ChemBERTa drug embedding + amino-acid composition; MLP	DAVIS training folds; six-fold unseen-target GroupKFold	paired with chembert_kiba2davis
4	ChemBERTa + AAC, KIBA to DAVIS	ChemBERTa drug embedding + amino-acid composition; MLP	KIBA; zero-shot transfer to DAVIS	paired with chembert_davisooof
5	ConPLex	protein language model + contrastive DTI score	pretrained on BindingDB; zero-shot DAVIS inference	possible training-data overlap with DAVIS proteins
8	Morgan + ESM-2, DAVIS OOF	Morgan drug features + ESM-2 target embedding; MLP	DAVIS training folds; six-fold unseen-target GroupKFold	paired with esm_target_kiba2davis
9	Morgan + ESM-2, KIBA to DAVIS	Morgan drug features + ESM-2 target embedding; MLP	KIBA; zero-shot transfer to DAVIS	paired with esm_target_davisooof
10	Morgan + AAC gradient boosting	Morgan + physicochemical drug features and amino-acid composition; HistGBM	DAVIS training folds; six-fold unseen-target GroupKFold	feature family overlaps several comparator arms
13	Morgan + AAC MLP, DAVIS OOF	Morgan + physicochemical drug features and amino-acid composition; MLP	DAVIS training folds; six-fold unseen-target GroupKFold	paired with mlp_kiba2davis
14	Morgan + AAC MLP, KIBA to DAVIS	Morgan + physicochemical drug features and amino-acid composition; MLP	KIBA; zero-shot transfer to DAVIS	paired with mlp_davisooof
16	Morgan + AAC random forest, DAVIS OOF	Morgan + physicochemical drug features and amino-acid composition; random forest	DAVIS training folds; six-fold unseen-target GroupKFold	paired with rf_kiba2davis
17	Morgan + AAC random forest, KIBA to DAVIS	Morgan + physicochemical drug features and amino-acid composition; random forest	KIBA; zero-shot transfer to DAVIS	paired with rf_davisooof

Table S8: Selection and estimator sensitivity of rank-selectivity associations. The 12-arm panel excludes eight chance-level arms on an outcome-related criterion. All coefficients are descriptive across dependent model arms.

Model panel	PR variable	endpoint	Pearson r	p	Spearman ρ	p	Kendall τ	p
12 above-chance	column-standardized PR	P@5	0.467	0.1261	0.347	0.2695	0.260	0.2426
12 above-chance	residual PR	P@5	-0.046	0.8877	-0.291	0.3593	-0.198	0.3716
12 above-chance	column-standardized PR	AUPRC	0.542	0.0685	0.462	0.1309	0.364	0.1160
12 above-chance	residual PR	AUPRC	-0.026	0.9354	-0.245	0.4433	-0.152	0.5452
all 20 scorers	column-standardized PR	P@5	0.309	0.1854	0.645	0.0021	0.483	0.0031
all 20 scorers	residual PR	P@5	0.039	0.8697	0.338	0.1452	0.243	0.1352
all 20 scorers	column-standardized PR	AUPRC	0.313	0.1791	0.660	0.0016	0.544	0.0008
all 20 scorers	residual PR	AUPRC	0.065	0.7866	0.373	0.1053	0.221	0.1859

For column-standardized PR versus P@5 in the 12-arm subset, the arm-resampling 95% interval was -0.39–0.89, leave-one-arm-out values ranged from 0.15 to 0.57, and the approximate two-sided 0.05 detectable absolute correlation was 0.58.

Table S9: Reusable spectral-audit protocol. Each step has a machine-readable implementation in the release. The generic command-line wrapper is `analysis/spectral_audit.py`; task-level validation remains dataset-specific.

Step	question / action	required output	release implementation
1	Define matrix and preprocessing	ligands, targets, missingness, clipping and scaling	<code>verify_inputs.py</code> ; <code>spectral_audit.py</code>
2	Separate estimands	column-standardized and two-way-centered residual surfaces	<code>centering_ladder</code>
3	Summarize concentration	PR, entropy rank, PC1 fraction, components for 90% variance	<code>rank_summary</code>
4	Quantify chemical-support uncertainty	paired ligand or scaffold-cluster intervals	<code>surface_bootstrap</code>
5	Probe target composition	leave-one-target-out and uniform/family-stratified subsampling	<code>target_jackknife</code> ; <code>target_subsampling</code>
6	Use transformation-matched nulls	parallel analysis, Gaussian additive and empirical residual nulls	<code>parallel_analysis</code> ; <code>additive_main_effect_null</code> ; <code>empirical_residual_permutation_null</code>
7	Check ranking invariance	decompose offsets, row effects and target-specific scales before task claims	<code>score-representation decomposition</code>
8	Validate operationally	absolute/transformed scores, target priors, identity/outcome permutations, assay and coverage strata	<code>expanded operational benchmark</code>
9	State the boundary	report uncertainty and equivalence status; do not equate dimension with accuracy	<code>manuscript interpretation rule</code>

Additional methodological notes

Interpretation of PR effective dimension. For a P -column standardized matrix with target correlation matrix C ,

$$r_{\text{PR}} = \frac{P^2}{\text{tr}(C^2)} = \frac{P}{1 + (P-1)\overline{r_{ij}^2}}.$$

The mean is over unique off-diagonal target correlations. The PR effective dimension therefore summarizes mean-squared inter-target correlation, while the plotted spectrum distinguishes a single dominant collective direction from several intermediate modes.

Limits of the experimental-noise calibration. The different-document repeatability scale includes technical variation, laboratory differences and assay-condition variation. It is deliberately conservative for a technical noise control, but it cannot represent target-specific systematic bias or all biological heterogeneity. The simulation therefore asks whether noise at the observed repeatability scale is sufficient to reproduce the experimental PR effective dimension; it is not a measurement-error correction of ChEMBL.

DTI model-panel interpretation. All 20 scorer arms are the primary exploratory panel in Table S6 and Fig. S5. The 12 arms in Table S7 share data, architectures and representations and were retained only as an explicitly outcome-restricted sensitivity. Table S8 compares both panels. The positive column-standardized-PR association in the all-20 panel and its absence in the restricted 12-arm panel show that the conclusion is selection- and estimator-dependent. In particular, `transformer_cnn_kiba2davis` had column-standardized PR dimension 8.21 at chance-level retrieval, while `morgan_cnn_kiba2davis` had residual PR dimension 31.75 at chance. High dimension is therefore not sufficient for useful selectivity, just as low dimension is not sufficient to diagnose failure. Reported correlations and nominal probabilities describe these panels only; they are not estimates of a population-level causal effect of dimensionality on selectivity.

For precision at five, we applied 20 iterations of Tukey-style median polish separately to predicted and experimental floor-excluded DAVIS matrices. Ligands with fewer than 12 co-observed targets were omitted. Experimental residuals at or above the within-ligand top decile, including ties, were labelled relevant; precision among the five largest predicted residuals was averaged over 68 eligible ligands. Eligible ligands had 13–402 measured targets (median 106), and 61 of 68 had an attainable precision ceiling of one.

Common-protocol parallel analysis. For each large docking matrix, the column-standardized and residual spectra were computed on the same 12,000-row subsample. Five hundred null spectra independently permuted each target column and then applied the corresponding transformation. The descriptive counts above the rank-wise 95th-percentile null were 3 versus 7 for Docking-44 and 4 versus 10 for DOCKSTRING-58. A studentized maximum-deviation envelope controlling the family-wise error rate over eigenvalue ranks retained the same counts. Each result was unchanged in five independent 100-permutation series. Because centering changes the estimand and creates one exact linear constraint, each surface is compared with its own transformed null.

Fitted additive null. The null retained the fitted grand, ligand and target effects and the observed target-specific residual variances, but independently sampled Gaussian residuals. On common 12,000-row supports, median residual PR dimensions were 42.40 and 56.48, compared with observed 9.29 and 18.16. The observed value was below all 500 realizations for each matrix (finite- simulation one-sided $p = 0.002$). This null shows that the empirical residual concentration is not a mechanical consequence of two-way centering.

Empirical residual-permutation null. Independently permuting every observed residual target column sampled exactly from each target's non-Gaussian empirical residual values, after which transformation-matched two-way centering and standardization were repeated. The latter step can slightly modify the marginals. Median residual PR dimensions were 42.40 for Docking-44 and 56.48 for DOCKSTRING-58; both observed values were below all 500 realizations ($p = 0.002$). Across 200 supports containing one ligand from each of 8,150 Docking-44 Butina clusters, observed residual PR had median 9.43 and the matched empirical null median 42.33. The empirical null does not retain ligand-specific residual scale, but the cluster-representative result shows that ordinary chemical redundancy does not explain the separation.

DOCKSTRING clipping and chemical support. The complete DOCKSTRING block contained 5,393 positive scores (0.0358%). Retaining rather than clipping them changed the column-standardized PR dimension from 2.266 to 2.275 and the residual value from 18.206 to 18.362. Across five independent 15,000-molecule supports, raw point estimates ranged from 2.252 to 2.294 and residual estimates from 18.161 to 18.322 (Fig. S6).

Matched target-preference sensitivities. The expanded primary analysis evaluated 2,522 non-tied observed target pairs from 137 ligands and 38 targets without imputing unobserved activities. At experimental margins of 0.10, 0.50 and 1.00 pChEMBL, 2,348, 1,779 and 1,178 pairs remained; residual Vina accuracies were 0.570, 0.569 and 0.586. Fifty-four absolute-Vina score ties received half credit. Across 38 leave-one-target-out deletions, residual accuracy ranged from 0.535 to 0.584 and the scaffold-held-out cohort prior from 0.583 to 0.627. Removing cells whose repeated values spanned more than one pChEMBL unit retained 118 ligands and 1,912 pairs, with residual accuracy 0.560. Human binding and human binding Ki/Kd restrictions gave 0.581 and 0.579. Before target-specific scaling, target-centered and two-way-centered scores differed by a constant within ligand and consequently produced exactly identical target rankings and accuracy 0.556. The fully residual-scaled score reached 0.564, so its operational difference is attributable to target-specific scale and its interaction with the ligand offset.

Both Murcko- and Butina-cluster intervals included zero for residual minus absolute and residual minus column-standardized scores. Conservative unions of the cluster-bootstrap 90% intervals were -0.056 – 0.051 and -0.015 – 0.049 , respectively. Neither comparison met a post hoc ± 0.02 equivalence margin; residual minus absolute also failed ± 0.05 . Holding the broad docking support fixed but comparing only within identical endpoint type retained 132 evaluable ligands and 4,330 endpoint-specific pair instances; absolute, column-standardized and residual accuracies were 0.553, 0.538 and 0.555. On human binding Ki/Kd measurements, 107 ligands supplied 2,224 pairs and the same accuracies were 0.565, 0.554 and 0.579. The residual-minus-absolute cluster intervals were -0.060 – 0.064 and -0.052 – 0.085 . These sensitivities remain observational and precision limited: target-pair co-observation had median support three, equivalence is not established, and no result demonstrates retrieval on unmeasured targets.